

Technology Stack

Date	13 July 2026
Team ID	-
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements. Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript, Jinja2	Creates a responsive and user-friendly web interface for students.
2	Backend / Server-Side	Python, FastAPI	Fast and efficient backend for AI request handling and API development.

3	Database / Data Storage	JSON Files (Local Storage)	Lightweight storage for project data, history, and feedback.
4	Cloud / Hosting / Deployment	Localhost (Development), GitHub Repository	Easy development, version management, and future cloud deployment.
5	Version Control & CI/CD	Git, GitHub	Source code management, collaboration, and version tracking.
6	Third-Party APIs / Other Tools	Google Gemini API, VS Code	Gemini API powers AI features like question answering, quizzes, recommendations, and explanations. VS Code is used for development.